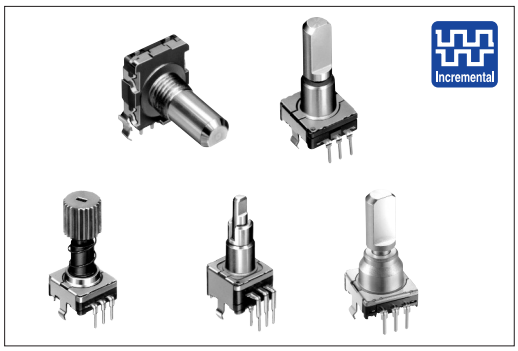
11mm Size Metal Shaft Type

EC11 Series

RoHS compliant

Compact and highly reliable type available in many varieties.



Typical Specifications

Items	Specifications
Rating	10mA 5V DC
Operating life	15,000cycles EC11EH:30,000cycles EC11K:100,000cycles 500,000cycles 1,000,000cycles
Operating temperature range	−40℃ to +85℃

Product Line

Structure	Actuator configuration	Actuator length (mm)	Torque (mN·m)	Number of detent	Number of pulse	Push-on switch	Travel of push-on switch (mm)	Operating life (cycles)	Minimum order unit (pcs.)	Product No.	Drawing No.
Horizontal	Flat	20	12±7	30	15	Without	——	15,000	1,400	EC11B15202AA	1
						With	0.5			EC11B15242AE	2
							1.5			EC11B15242AF	3
Vertical			10±7	18	9	Without	——		2,400	EC11E09204A4	4
				30	15					EC11E15204A3	
			7^{+3}_{-4}	Without						EC11E1530401	
			10±7	36	18					EC11E1820402	
			7^{+3}_{-4}	Without						EC11E1830401	
			10±7	18	9	With	0.5			EC11E09244BS	5
				30	15					EC11E15244G1	
			7^{+3}_{-4}	Without						EC11E153440D	
			10±7	36	18					EC11E18244AU	
			7^{+3}_{-4}	Without						EC11E183440C	
			10±7	18	9		1.5			EC11E09244AQ	
				30	15					EC11E15244B2	
			7^{+3}_{-4}	Without						EC11E1534408	
			10±7	36	18					EC11E18244A5	
			7^{+3}_{-4}	Without						EC11E1834403	
			12±5	18	9	Without	——	100,000	2,000	EC11K0920401	6
				30	15					EC11K1520401	
				18	9	With	0.5			EC11K0924401	7
				30	15					EC11K1524402	
				18	9		1.5			EC11K0925401	
				30	15					EC11K1525401	
				18	9	Without	——	500,000		EC11K0920601	6
				30	15					EC11K1520601	
				18	9	With	0.5			EC11K0924601	7
				30	15					EC11K1524601	
				18	9		1.5			EC11K0925601	
				30	15					EC11K1525601	
				18	9	Without	——	1,000,000		EC11K0920801	6
				30	15					EC11K1520801	
				18	9	With	0.5			EC11K0924801	7
				30	15					EC11K1524801	
				18	9		1.5			EC11K0925801	
				30	15					EC11K1525801	

Product Line

Structure	Actuator configuration	Actuator length (mm)	Torque (mN·m)	Number of detent	Number of pulse	Push-on switch	Travel of push-on switch (mm)	Operating life (cycles)	Minimum order unit (pcs.)	Product No.	Drawing No.
Push lock	20-tooth serration	25	10±7	30	15	Without	——	15,000	1,600	EC11E152T409	8
		26.4				With	8			EC11E152U402	9
Self-return switch	Flat	15	3 to 30	Without	Self-return switch	Without	——		2,400	EC1110120005	10
		20				With	0.5			EC111012010H	11
							1.5			EC1110120201	
ABC switch			10±7	18	ABC switch		0.5	30,000		EC11EH124403	12
				30						EC11EH224403	
				18			1.5			EC11EH124404	
				30						EC11EH224404	
Dual-shaft	Slotted	Inner-shaft=25			15		0.5	15,000	1,400	EC11EBB24C03	13
		Outer-shaft=15				Without	—				
	Flat	Inner-shaft=25				With	1.5			EC11E0B2LB01	14
	Slotted	Outer-shaft=15	3 to 30	Without	Self-return switch	Without	—				

Packing Specifications

Tray

Product No.	Number of packages (pcs.)		Export package measurements (mm)
	1 case /Japan	1 case /export packing	
EC11B	700	1,400	370×520×201
EC11E09/15/18/EC11EH	1,200	2,400	540×360×250
EC11K	1,000	2,000	
EC11E152T/U	800	1,600	374×529×213
EC111	1,200	2,400	363×507×216
EC11E0B/BB	700	1,400	374×529×213

Notes

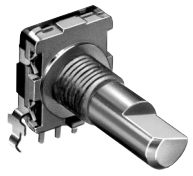
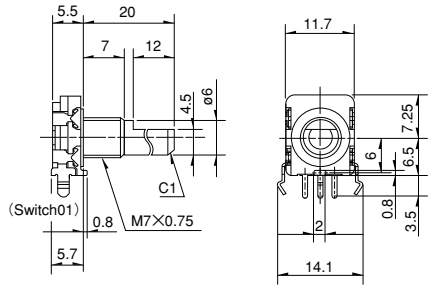
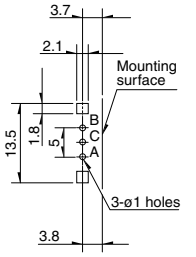
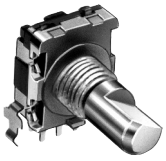
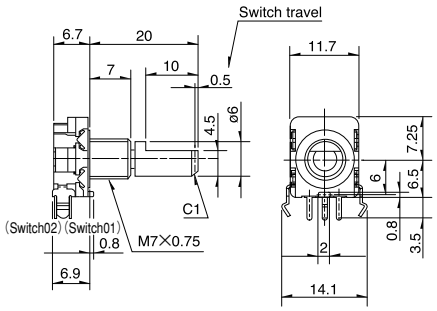
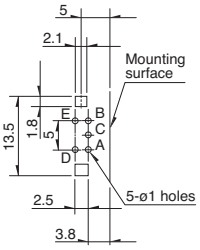
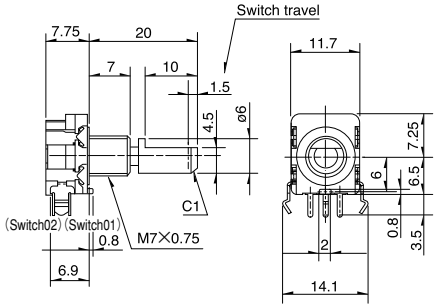
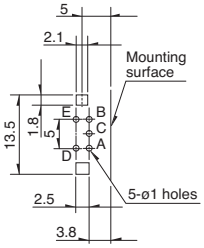
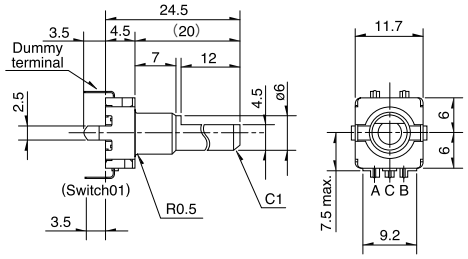
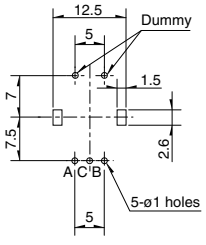
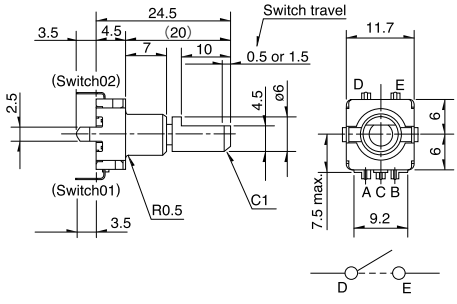
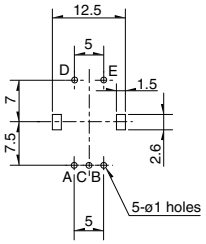
1. Products other than those listed in the above chart are also available. Please contact us for details.
2. Please place purchase orders per minimum order unit N (integer).

Automotive Use

Refer to **P.209** for product varieties.
Refer to **P.217** for product specifications.
Refer to **P.217** for attached parts.

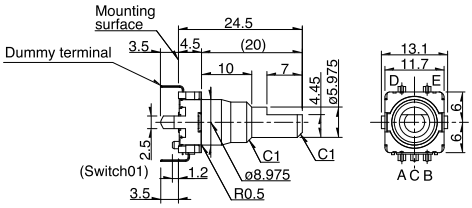
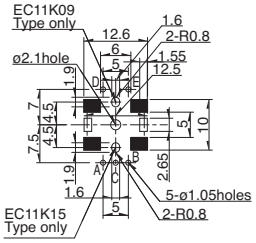
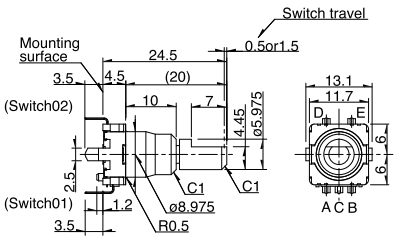
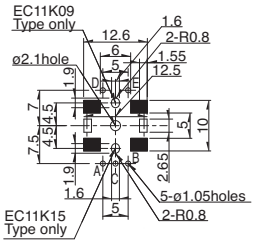
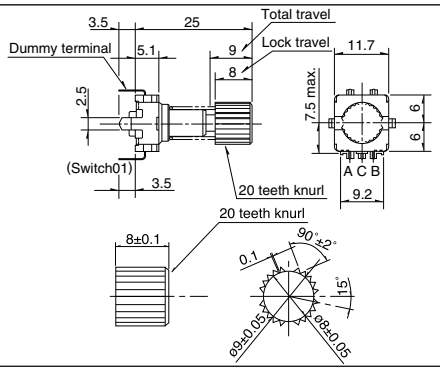
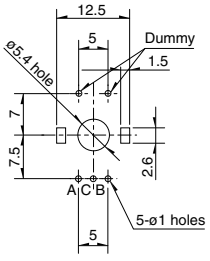
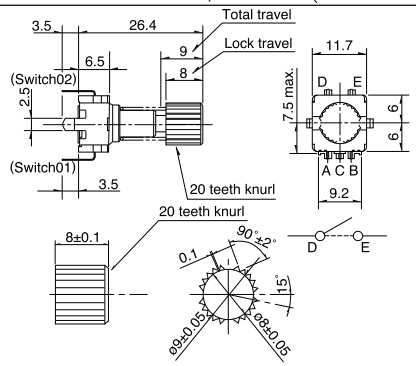
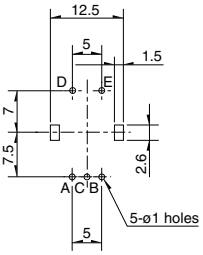
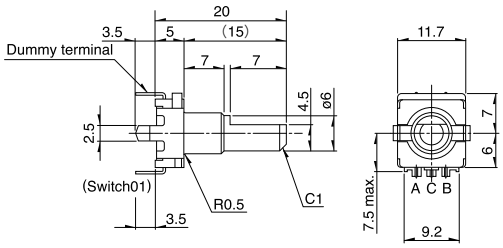
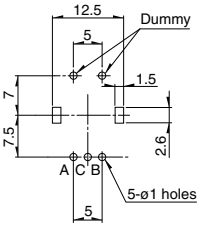
Dimensions

Unit:mm

No.	Photo	Style	PC board mounting hole dimensions (Viewed from mounting side)
1	Horizontal 		
2	Horizontal with push-on switch (travel 0.5mm) 		
3	Horizontal with push-on switch (travel 1.5mm) 		
4	Vertical 		
5	Vertical with push-on switch (travel 0.5/1.5mm) 		

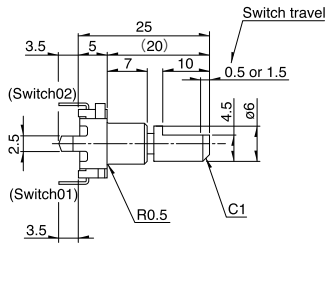
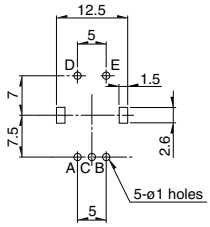
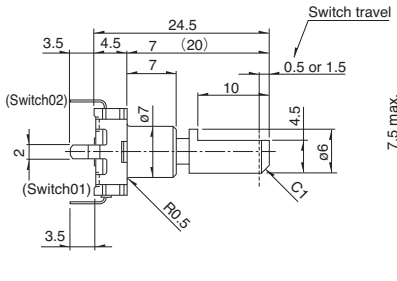
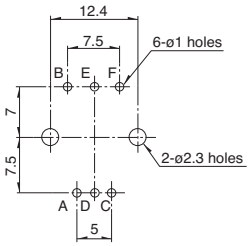
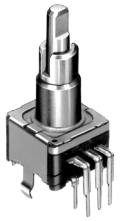
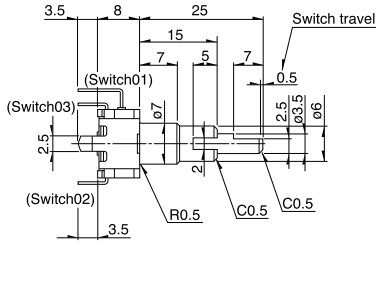
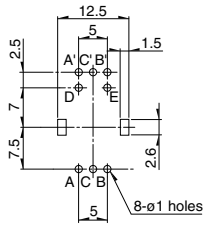
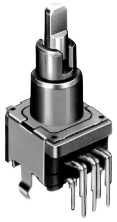
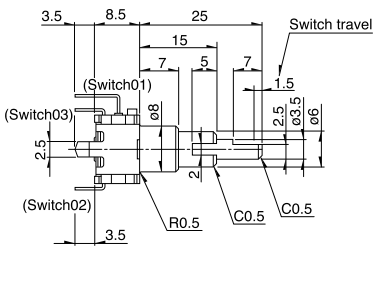
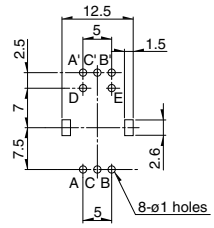
Dimensions

Unit:mm

No.	Photo	Style	PC board mounting hole dimensions (Viewed from mounting side)
6	Vertical 		 PC board mounting detail A slant line part hows the solder land black part: Do not solder and wiring for electrical contact
7	Vertical with push-on switch (travel 0.5mm/1.5mm) 		 PC board mounting detail A slant line part hows the solder land black part: Do not solder and wiring for electrical contact
8	Push-lock mechanism 		
9	Push-lock mechanism with push-lock switch 		
10	Self-return switch 		

Dimensions

Unit:mm

No.	Photo	Style	PC board mounting hole dimensions (Viewed from mounting side)
11	Self-return switch with push-on switch (travel 0.5mm/1.5mm) 		
12	ABC switch with push-on switch (travel 0.5mm/1.5mm) 		
13	Dual-shaft type with push-on switch (travel 0.5mm) Inner shaft: encoder Outer shaft: encoder 		
14	Dual-shaft type with push-on switch (travel 1.5mm) Inner shaft: encoder Outer shaft: self-return switch 		

Product Varieties

Shaft Dimensions

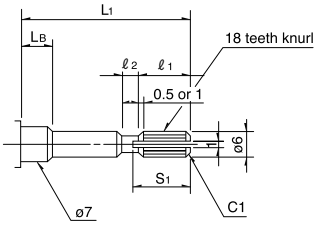
1. Single-shaft Type (EC11K is excluded. Refer to P.212 for EC11K)

1) Knurled Type

Unit:mm

Configuration (Shaft diameter: φ6)

Not applicable for EC11E with push-lock mechanism



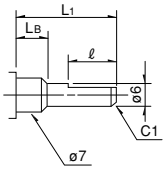
Detailed dimensions

L1	LB	ℓ1	ℓ2	S1
20	7	6	1	7
25	10	10	2	11

2) Flat Type

Unit:mm

Configuration (Shaft diameter: φ6)



Detailed dimensions

	L1	LB	ℓ
※	15	5	7
	15	7	5 (6)
	20	7	10 (12)
	25	10	12

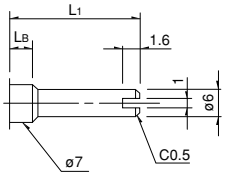
※Does not comply with EC111

Dimensions marked with ( ) apply to products without push-on switches.

3) Slotted Type

Unit:mm

Configuration (Shaft diameter: φ6)



Detailed dimensions

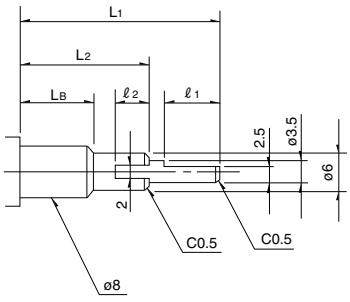
L1	LB
15	7
20	7
25	10

2. Standard Dimensions and configuration of Dual-shaft Type

1) Flat Type

Unit:mm

Configuration (Inner-shaft : φ3.5 Outer-shaft : φ6)



Detailed dimensions

L1	L2	LB	ℓ1	ℓ2
20	10	5	7	4
25	15	7	7	5
30	20	10	7	5

Notes

1. The highlighted figures in shaft types refer to Product Specifications in P.204 and P.205.
2. Products other than those listed in the above chart are also available. Please contact us for details

List of Varieties

Type		Metal shaft									
		9mm size	11mm size								
Series		EC09E	EC11B	EC11E		EC111		EC11EH			
Photo											
Output		Incremental				Self-return switch		ABC switch			
Outline specifications	Shaft types	Single-shaft			Dual-shaft	Single-shaft					
	Operating direction	Vertical	Horizontal	Vertical							
	Number of pulse/ Number of detent	15/30		9/18 15/30 18/36		———		6×ABC/18 10×ABC/30			
	Push switch (Travel mm)	Without/With (0.5, 1.5)									
	Optional functions	———		Push-lock mechanism Without detent		———					
	Changeover angle	———				17 to 40°		———			
Dimensions (mm)	W	9.5	11.7								
	D		13.8	12		13		12			
	H	4.5	5.5	4.5	8.5	5		4.5			
Soldering	Manual soldering	350℃ max. 3s max.									
	Dip soldering	260±5℃, 3±1s	260±5℃, 5±1s								
Operating temperature range		−40℃ to +85℃									
Automotive use		○	○	○		○		○			
Maximum operating current (Resistive load)		10mA									
Electrical performance	Output signal	Output of A and B signals, proportionate to phase difference				Self-return switch		Three phase A, B and C			
	Output wave	Please see P.217									
	Sliding noise	Please see P.217									
	Insulation resistance	100MΩ min. 250V DC									
	Voltage proof	300V AC for 1minute									
Mechanical performance	Rotational torque	———		7^{+3}_{-4} mN·m	—	3～30mN·m		———			
	Detent torque	8±5mN·m	12±7mN·m	10±7mN·m		———		10±7mN·m			
	Push-pull strength	100N									
Environmental performance	Cold	−40±3℃ for 240h									
	Dry heat	+85±3℃ for 240h									
	Damp heat	60±2℃, 90 to 95%RH for 240h									
Page		203	204								

Push-on Switch Specifications

Items	EC09E		EC11B		EC11E		EC111		EC11EH	
Contact arrangement	Single pole and single throw (Push-on)									
Travel (mm)	0.5±0.3	1.5±0.5	$0.5^{+0.4}_{-0.3}$	1.5±0.5	0.5±0.3	1.5±0.5 8±0.8※	0.5±0.3	1.5±0.5	0.5±0.3	1.5±0.5
Operating force (N)	$6^{+2.5}_{-2}$	4±2	6±3	5±2	$6^{+2.5}_{-2}$	4±2 8 max. ※	$6^{+2.5}_{-2}$	4±2	$6^{+2.5}_{-2}$	4±2
Rating	10mA 5V DC		0.1A 5V DC (500μA 5V DC min. ratings)		0.1A 5V DC (500μA 5V DC min. ratings) 0.5A 12V DC ※		0.1A 5V DC (500μA 5V DC min. ratings)			
Contact resistance	100mΩ max. for initial period; 200mΩ max. after operating life.									
Operating life	10,000times min.		25,000 times min.	20,000 times min.	20,000times min. 10,000times min.※		20,000times min.			

Notes

1. ※marked specification is only applicable to EC11E152U402.
2. ○marks in "Available for automotive use" indicate that some of the series products can work at the operating temperature range from -40℃ to +85℃.

List of Varieties

Type		Metal shaft			
		11mm size		20mm size	
Series		EC11K	EC11J	EC20A/RK203	EM20B
Photo					
Output		Incremental			
Outline specifications	Shaft types	Single-shaft			
	Operating direction	Vertical			
	Number of pulse/ Number of detent	9/18 15/30		18/18	40/40
	Push switch (Travel mm)	Without With (0.5, 1.5)		Without With (1.5)	Without With(0.5)
	Optional function	————			Display function
Dimensions (mm)	W	11.7		20.2	20
	D	12	14.2	19.2	22.25
	H	4.5		10	13
Soldering	Manual soldering	350℃ max. 3s max.			
	Dip soldering	260±5℃, 3±1s	————	260±5℃, 5±1s	260℃ max. 3s max.
	Reflow soldering	————	Please see P.212	————	
Operating temperature range		−40℃ to +85℃			−10℃ to +70℃
Automotive use		○	○	○	——
Maximum operating current (Resistive load)		10mA			——
Electrical performance	Output signal	Output of A and B signals, proportionate to phase difference			V_{HI} : +4.0V min. V_{LO} : 0.5V max. (Isink=5mA)
	Output wave	Please see P.217			——
	Output phase	————			A, B (Squarewave)
	Sliding noise	Please see P.217			——
	Insulation resistance	100MΩ min. 250V DC		10MΩ min. 50V DC	100MΩ min. 250V DC
	Voltage proof	300V AC for 1minute		50V AC for 1minute	300V AC for 1minute
Mechanical performance	Rotational torque	————			7mN·m max.
	Detent torque	12±5mN·m		40±20mN·m	8±5mN·m
	Push-pull strength	100N			
Environmental performance	Cold	−40±2℃ for 240h		−40±3℃ for 240h	−30±3℃ for 240h
	Dry heat	85±2℃ for 240h		85±3℃ for 240h	80±3℃ for 240h
	Damp heat	60±2℃, 90 to 95% RH for 240h		60±3℃, 90 to 95%RH for 240h	40±2℃, 90 to 95%RH for 240h
Page		204	210	213	215

Push-on Switch Specifications

Items	EC11K/EC11J		EC20A	EM20B
Contact arrangement	Single pole and single throw (Push-on)			
Travel (mm)	0.5±0.3	1.5±0.5		$0.5^{+0.4}_{-0.3}$
Operating force (N)	5±2	4±2		6±3
Rating	0.1A 5V DC		0.5A 16V DC (1mA 16V DC min. ratings)	0.1A 5V DC
Contact resistance	100mΩ for initial period; 200mΩ after operating life			
Operating life	1,000,000 times min.	100,000 times min.	20,000times min.	25,000times min.

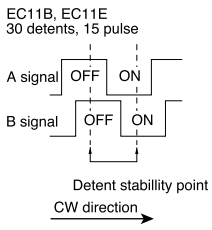
Note

○marks in "Available for automotive use" indicate that some of the series products can work at the operating temperature range from -40℃ to +85℃.

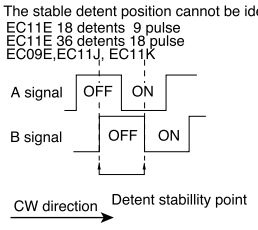
Product Specifications

Output Wave

EC11B

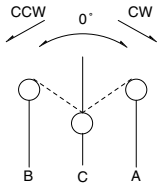


EC09E/EC11E/EC11J/EC11K

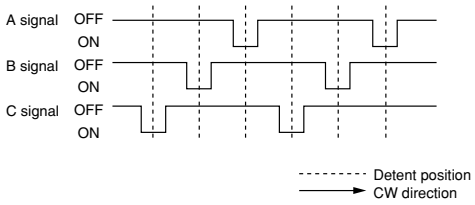


Circuit Diagram

EC111/EC11E0B



EC11EH



EC20A

Shaft rotational Direction	Signal	Output
Clockwise	A (Terminal A–C)	OFF / ON
	B (Terminal B–C)	OFF / ON
Counter-clockwise	A (Terminal A–C)	OFF / ON
	B (Terminal B–C)	OFF / ON

The dotted lines indicate detent positions

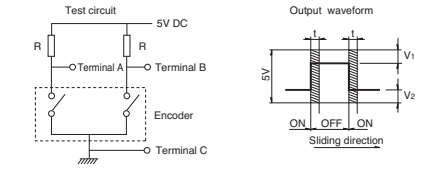
Sliding Noise

EC09E

$V_1 = V_2 = 1.5V$ max.

At R = 5kΩ
Chattering: 5ms max.
Bounce: 2ms max.

EC11B

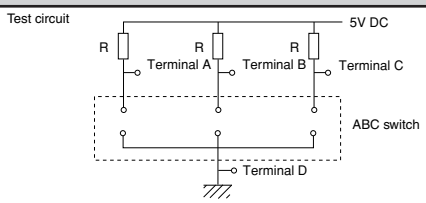


At R = 5kΩ
Chattering : 2ms max.
Bounce : 2ms max.

EC11E

At R = 5kΩ
Chattering : 3ms max.
Bounce : 2ms max.

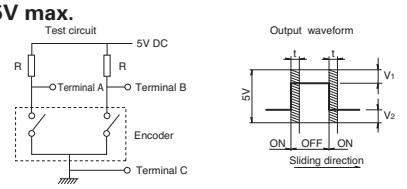
EC11EH



At R = 5kΩ
Chattering: 5ms max.
Bounce : 5ms max.

EC20A

$V_1 = V_2 = 2.5V$ max.



At R = 5kΩ
Chattering: 8ms max.
Bounce: 5ms max.

EC11J/EC11K

At R = 5kΩ
Chattering : 3ms max.
Bounce : 2ms max.

Attached Parts

EC11B Series

Unit:mm

